

Three Latency Metrics — Measurement Scope Clarification

Paper claims:

'>30x vs ViT4Mal 810 ms' → use Full Application Latency (43.7x)

'Q/K pruning reduces HW 9%' → use FPGA Kernel Latency (-9.3%)

DO NOT use Kernel-only time for ViT4Mal speedup (gives 800x, misleading)

Latency (log scale)

10^6

10^5

10^4

10^3

ViT4Mal
fastest [2]

QKS-v2 S0
(Full App)

QKS-v2 S20
(Full App)

QKS-v2 S0
(Kernel only)

QKS-v2 S30
(Kernel only)

18.5 ms

18.6 ms

1004 μ s

911 μ s

Scope: JPEG→padding→FPGA→output
= our Full Application Latency

Scope: FPGA IP execution only
= proves HW sparse acceleration